

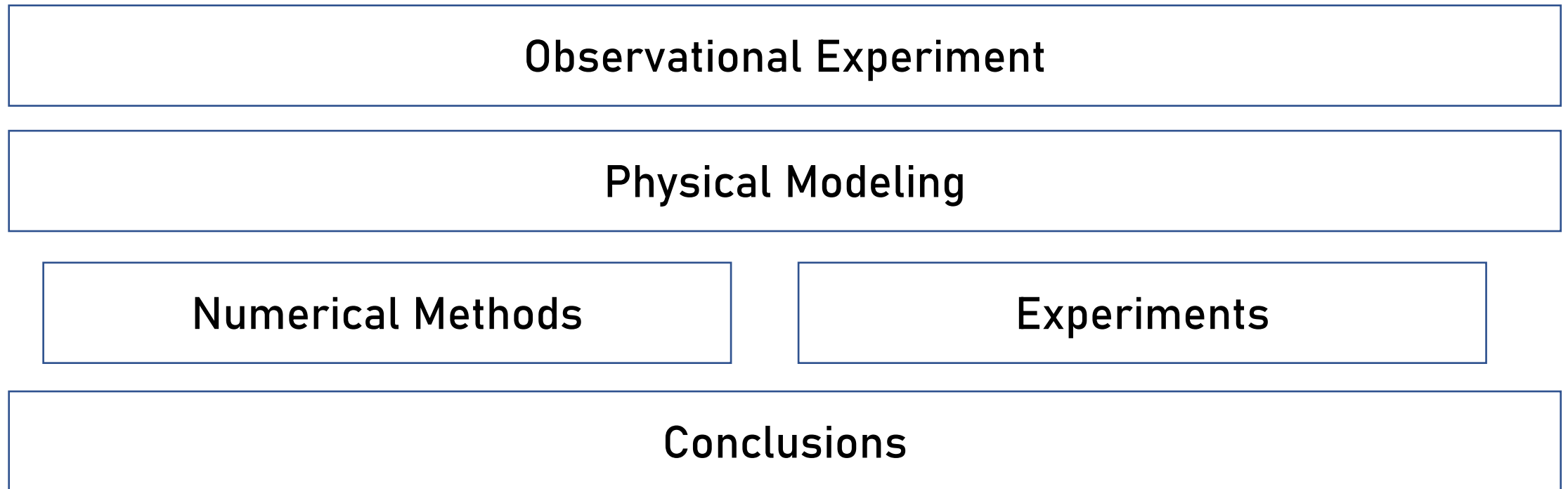
14. Looping Pendulum

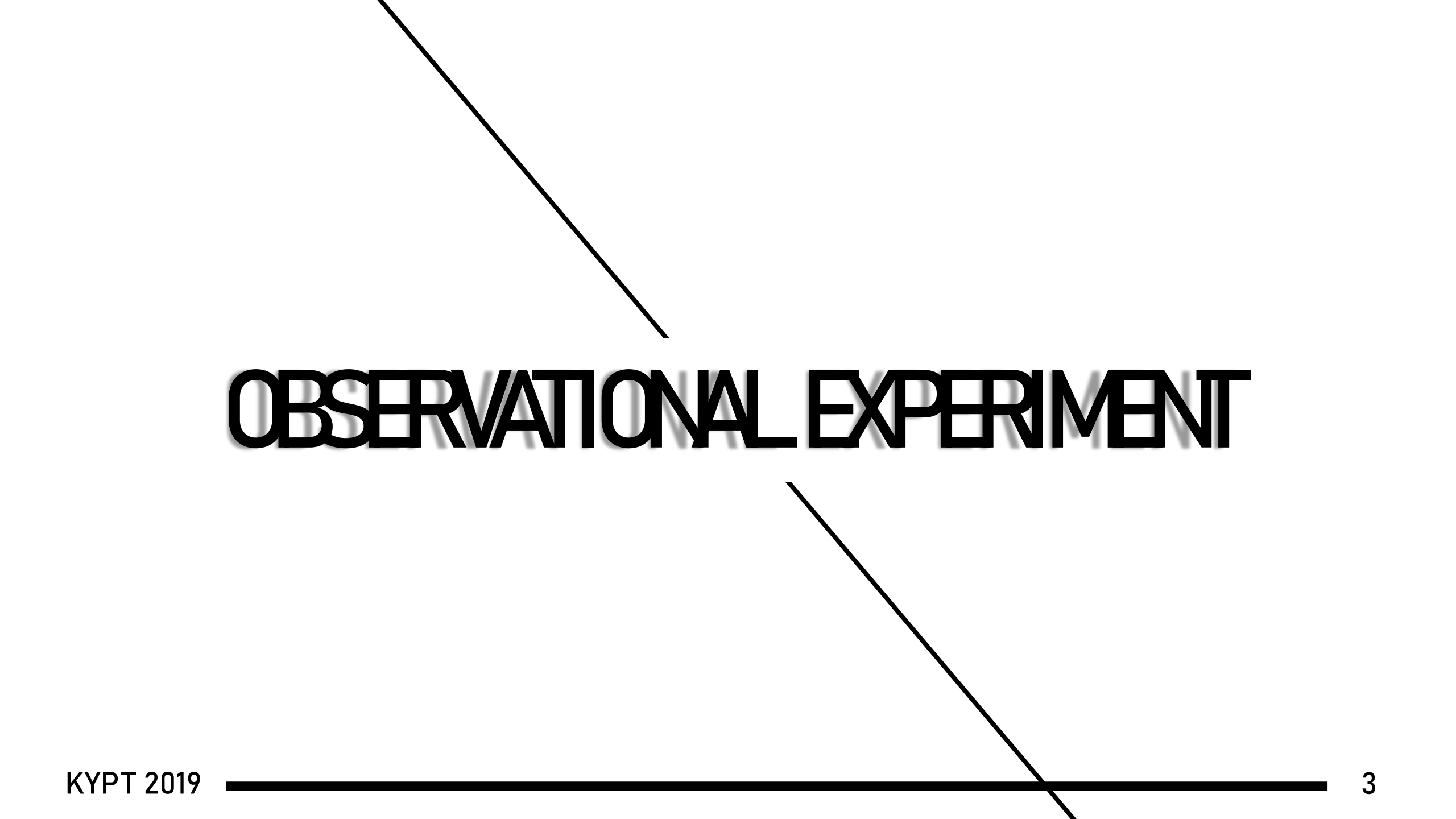
HyunJun Lee | 5minsleft

Connect two loads, one heavy and one light, with a string over a horizontal rod and lift up the heavy load by pulling down the light one. Release the light load and it will sweep around the rod, keeping the heavy load from falling to the ground. Investigate this phenomenon.



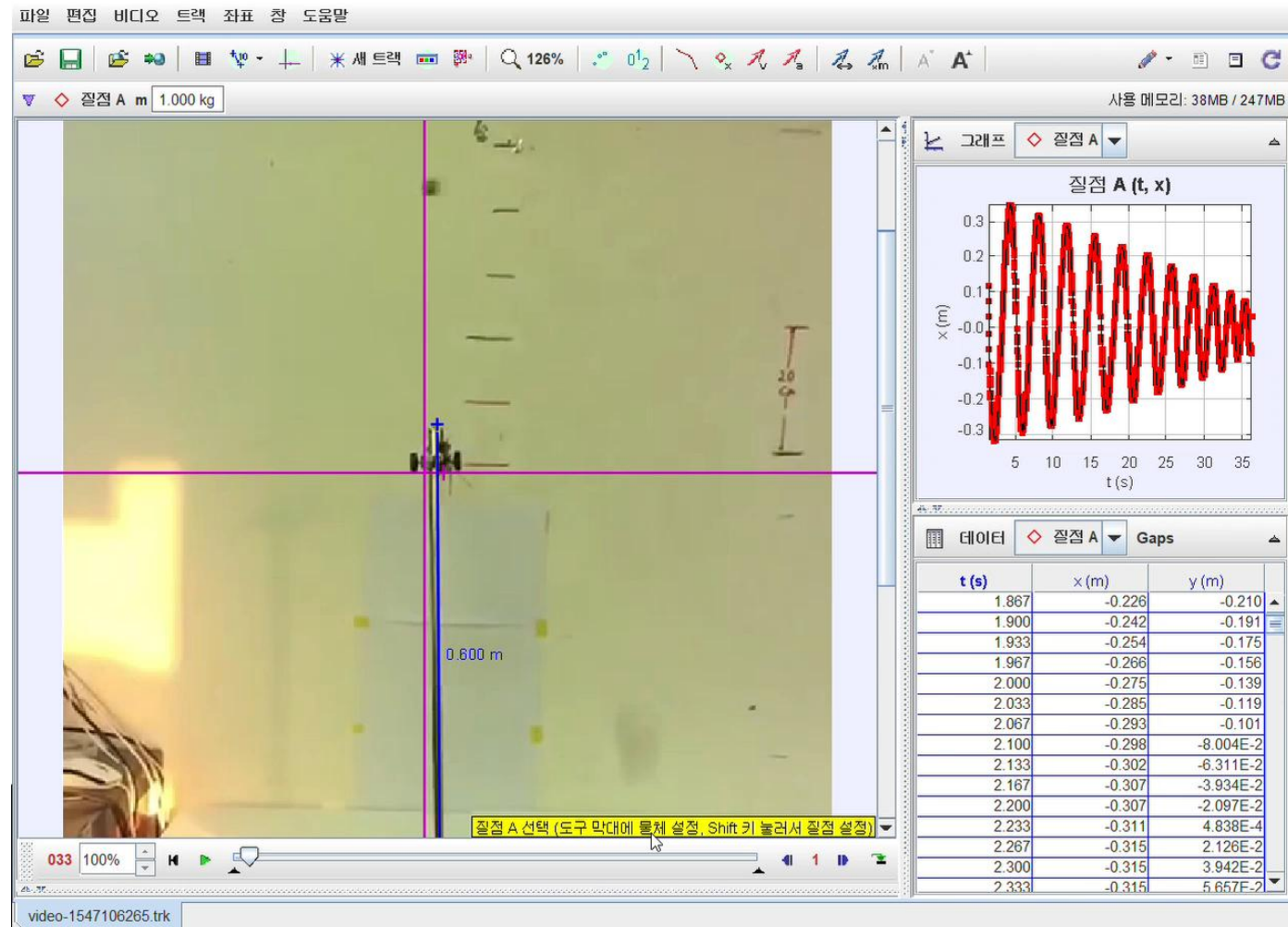
FLOWCHART





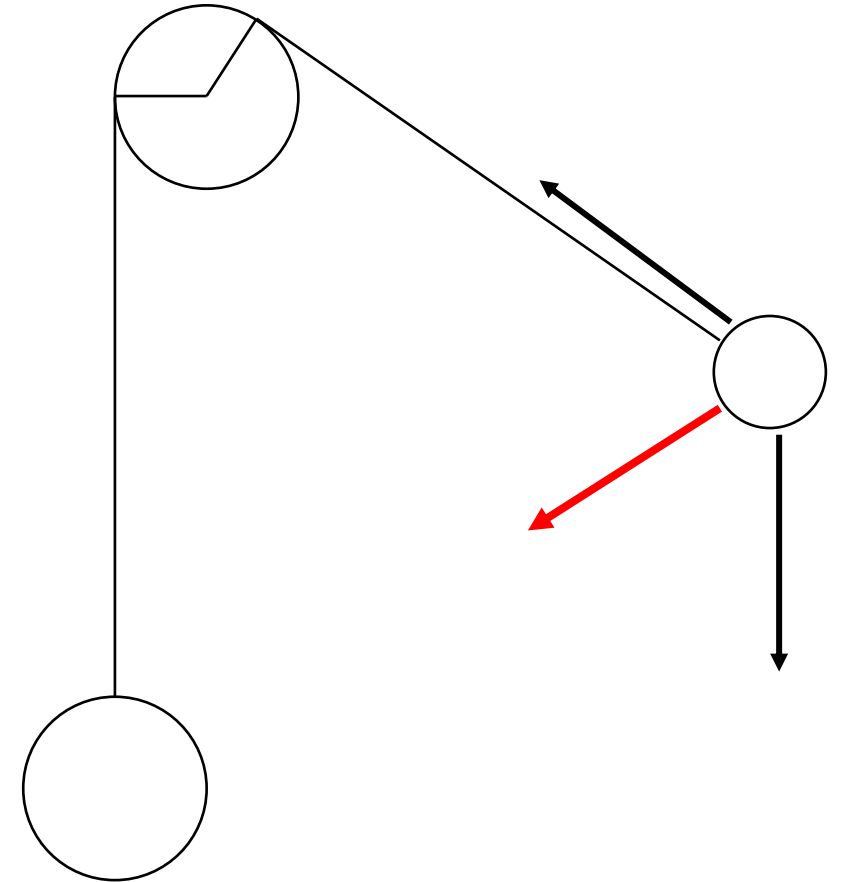
OBSERVATIONAL EXPERIMENT

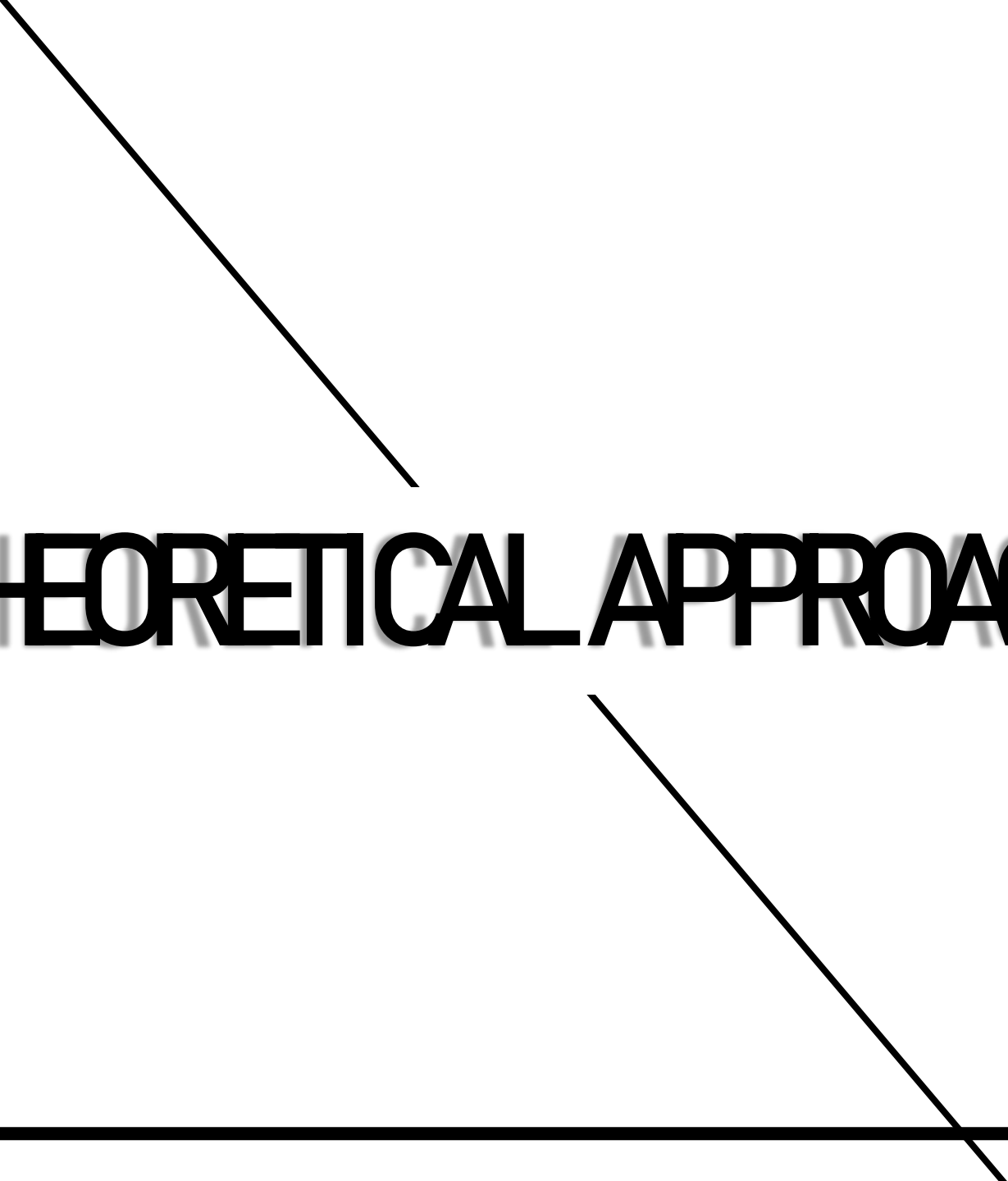
OBSERVATIONAL EXPERIMENT



QUALITATIVE EXPLANATION

1. The lighter object starts to move because of the tension made by the heavier object.
2. As the lighter object moves around the rod, the tension on the string connected to the heavier object increases, decelerating the heavier object.
3. When the heavier object stops, static friction acts on the string, after which the string does not slip around the rod.
4. The motion of the lighter object is then independent from the heavier one.



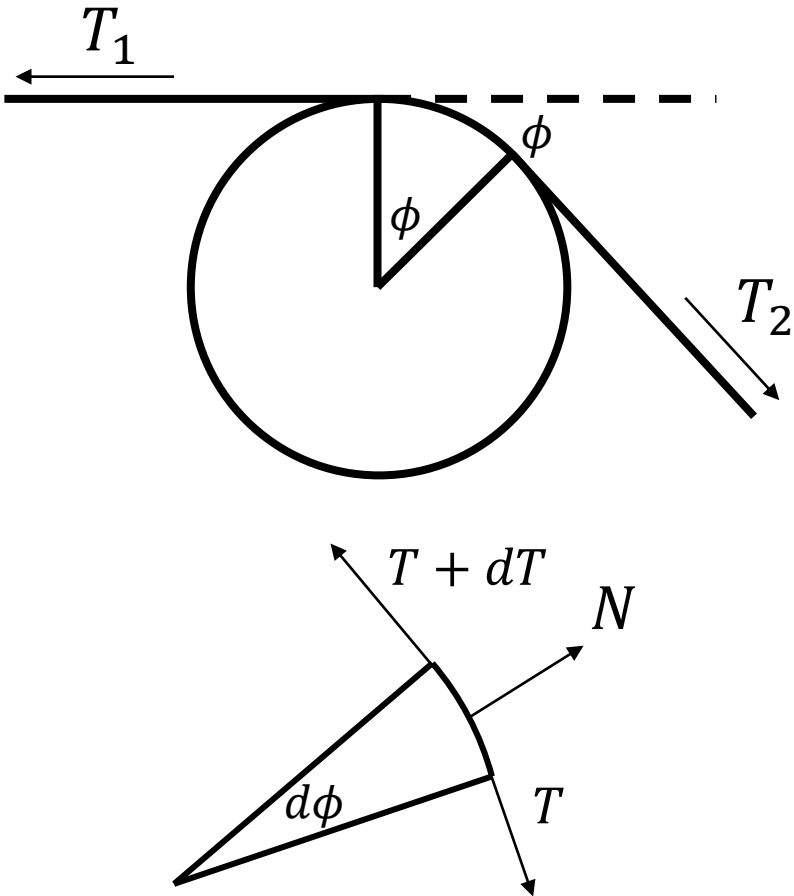


THEORETICAL APPROACH

THEORY_ ASSUMPTIONS

1. Two-dimensional movement
2. The loads are point masses
3. Constant friction factor between string and rod
4. Air resistance is negligible

THEORY_CAPSTAN EQUATION



Normal Force

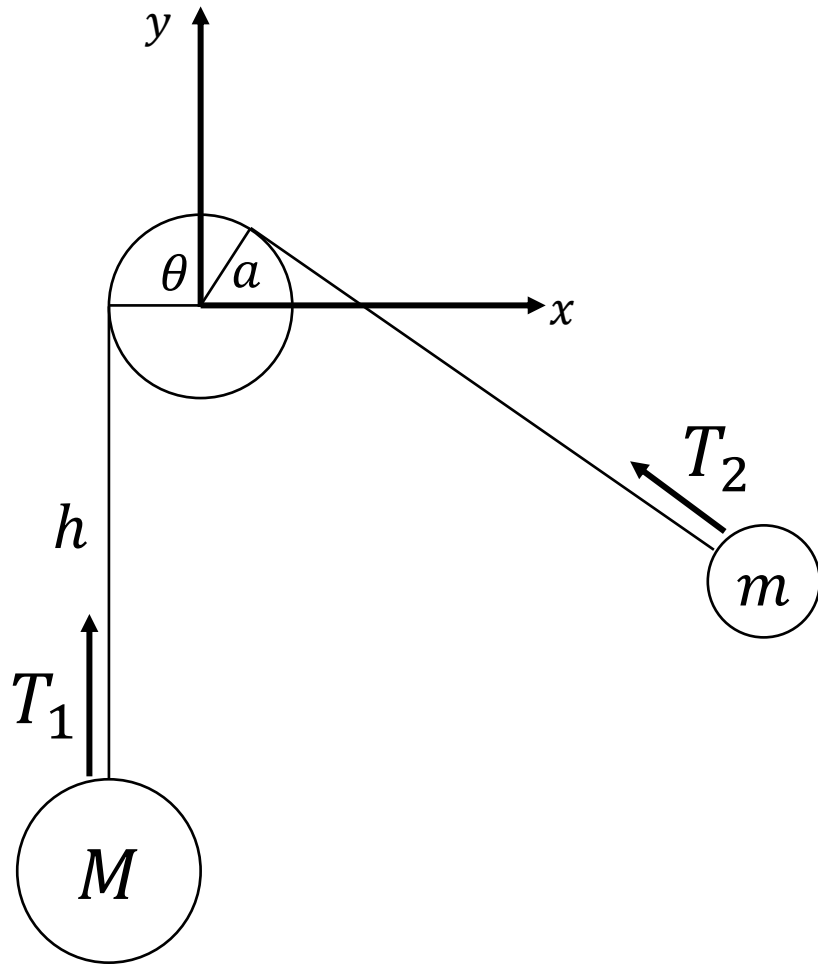
$$N = (T + (T + dT)) \sin \frac{d\phi}{2} = T d\phi$$

Increase in Rope Tension

$$dT = \mu N$$

$$\int \frac{1}{T} dT = \int \mu d\phi \longrightarrow \boxed{T_1 = T_2 e^{\mu\phi}}$$

THEORY_COORDINATES



Coordinates of m

$$x = -a \cos \theta + (L - h - a \theta) \sin \theta$$

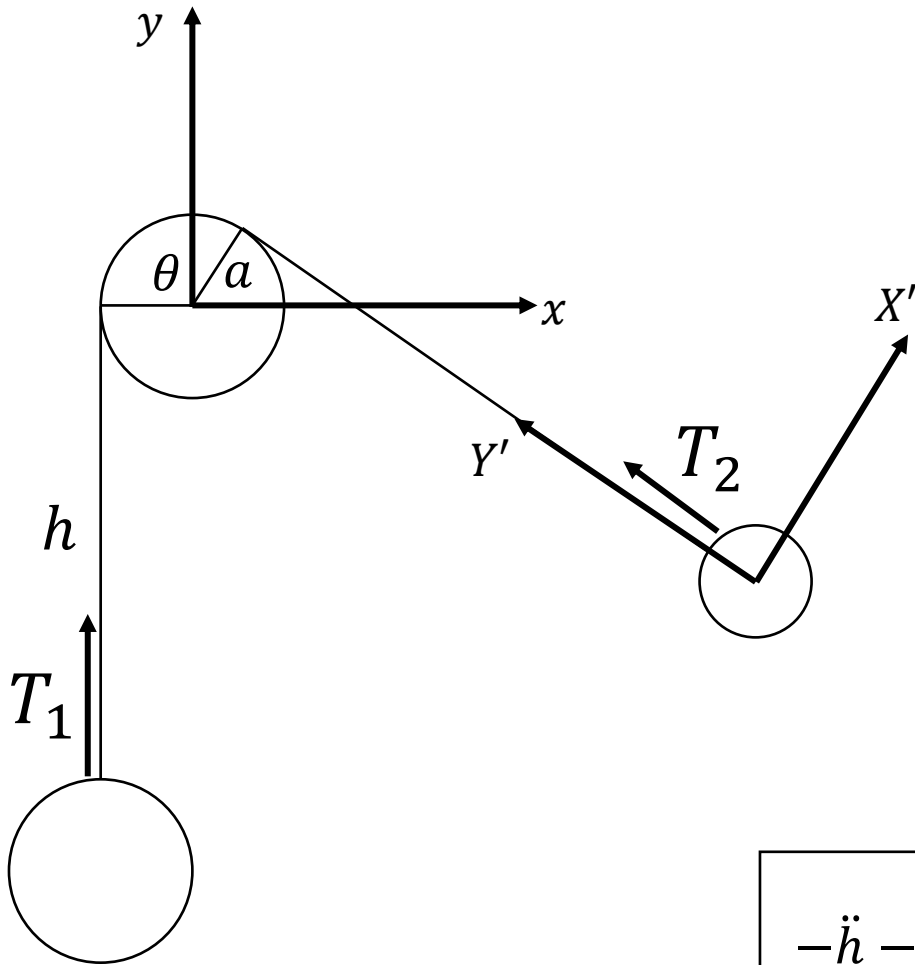
$$y = (L - h - a \theta) \cos \theta + a \sin \theta$$

Differentiation by time

$$a_x = -\ddot{h} \sin \theta - 2 \dot{h} \dot{\theta} \cos \theta + L \ddot{\theta} \cos \theta - L \dot{\theta}^2 \sin \theta - h \ddot{\theta} \cos \theta + h \dot{\theta}^2 \sin \theta - a \dot{\theta}^2 \cos \theta - a \theta \ddot{\theta} \cos \theta + a \theta \dot{\theta}^2 \sin \theta$$

$$a_y = -\ddot{h} \cos \theta + 2 \dot{h} \dot{\theta} \sin \theta - L \ddot{\theta} \sin \theta - L \dot{\theta}^2 \cos \theta + h \ddot{\theta} \sin \theta + h \dot{\theta}^2 \cos \theta + a \dot{\theta}^2 \sin \theta + a \theta \ddot{\theta} \sin \theta + a \theta \dot{\theta}^2 \cos \theta$$

THEORY_ GOVERNING EQUATION



Coordinates of m

$$\ddot{X}' = -2 \dot{h} \dot{\theta} + L \ddot{\theta} - h \ddot{\theta} - a \dot{\theta}^2 - a \theta \ddot{\theta} = g \sin \theta$$

$$\ddot{Y}' = -\ddot{h} - L \dot{\theta}^2 + h \dot{\theta}^2 + a \theta \dot{\theta}^2 = -g \cos \theta - \frac{T_2}{m}$$

Equations of Motion about M

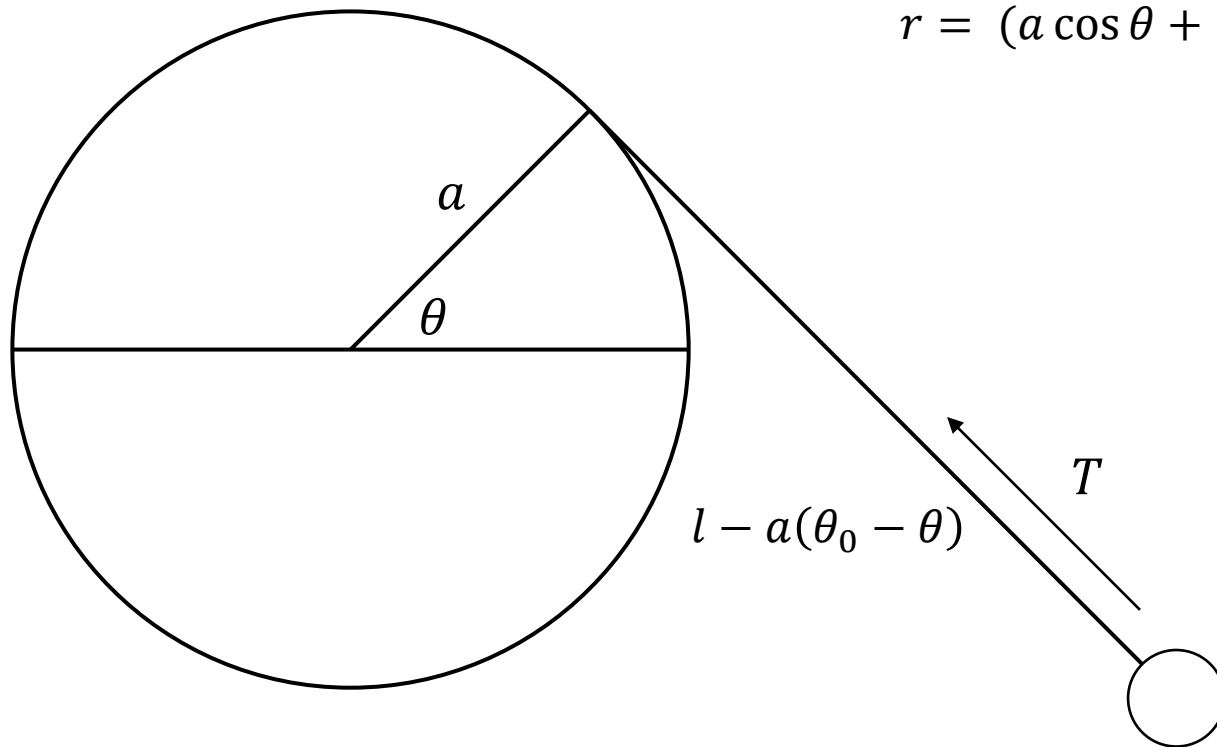
$$M \ddot{h} = -T_1 + Mg$$

Capstan Equation

$$T_2 = T_1 e^{-\mu_k \theta}$$

$$-\ddot{h} - L \dot{\theta}^2 + h \dot{\theta}^2 + a \theta \dot{\theta}^2 = -g \cos \theta - \frac{M}{m} e^{-\mu_k} (g - \ddot{h})$$

THEORY_ ENERGY CONSERVATION



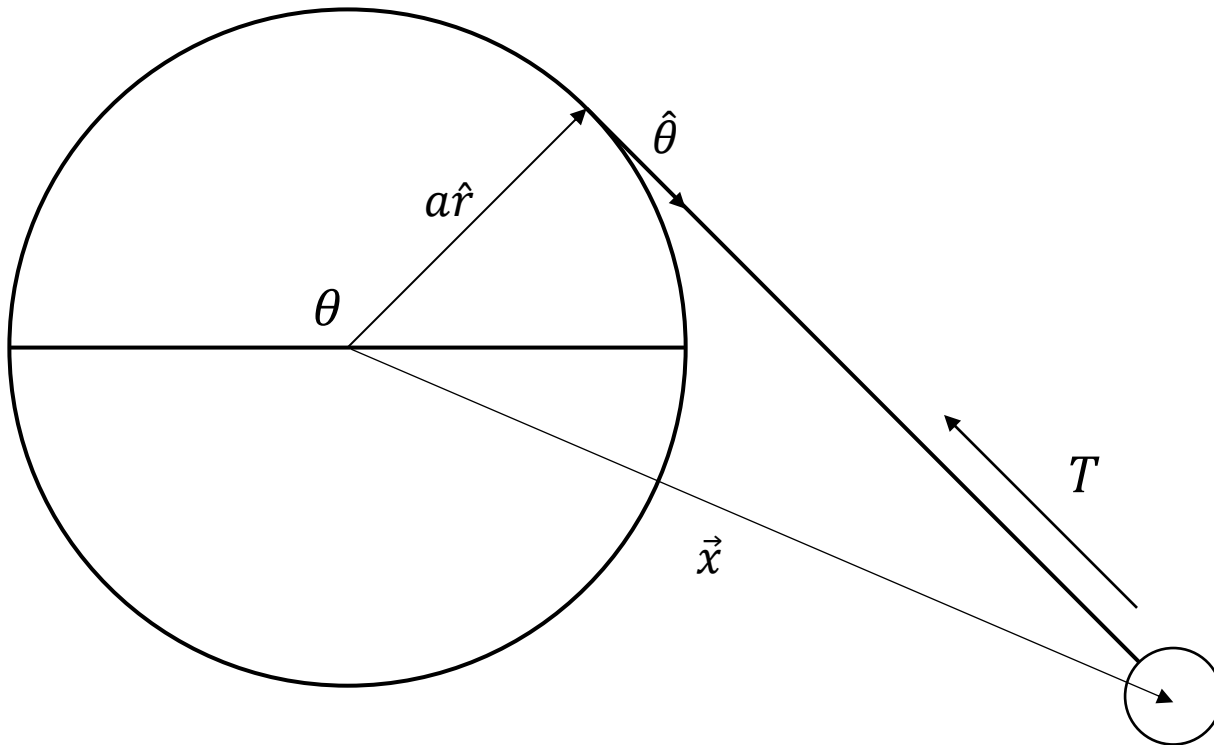
$$r = (a \cos \theta + (l - a(\theta_0 - \theta)) \sin \theta, a \sin \theta - (l - a(\theta_0 - \theta)) \cos \theta)$$

$$\dot{r} = ((l - a\theta_0) \cos \theta, (l - a\theta_0) \sin \theta)$$

$$T = T(-\sin \theta, \cos \theta)$$

$$T \cdot \dot{r} = 0$$

THEORY_ ENERGY CONSERVATION

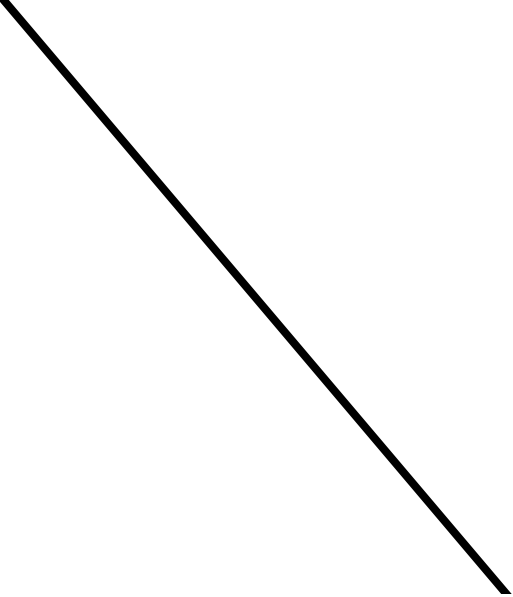


$$\vec{x} = a\hat{r} + (L - a\theta)\hat{\theta}$$

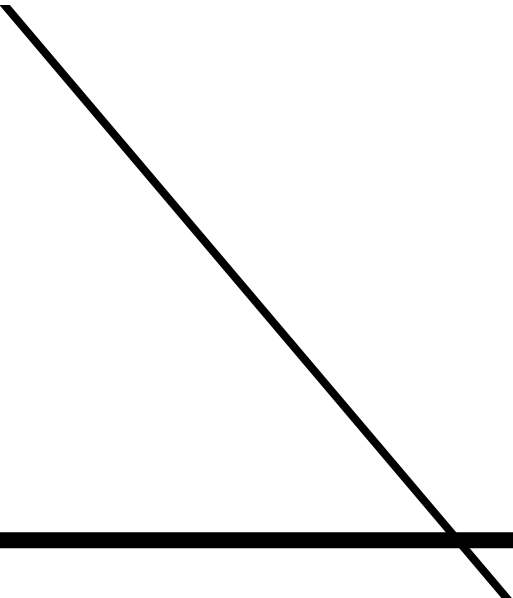
$$\begin{aligned}\dot{\vec{x}} &= a\dot{\theta}\hat{\theta} - a\dot{\theta}\hat{\theta} - (L - a\theta)\dot{\theta}\hat{r} \\ &= -(L - a\theta)\dot{\theta}\hat{r}\end{aligned}$$

$$\vec{T} = -T\hat{\theta}$$

$$\vec{T} \perp \dot{\vec{x}}$$

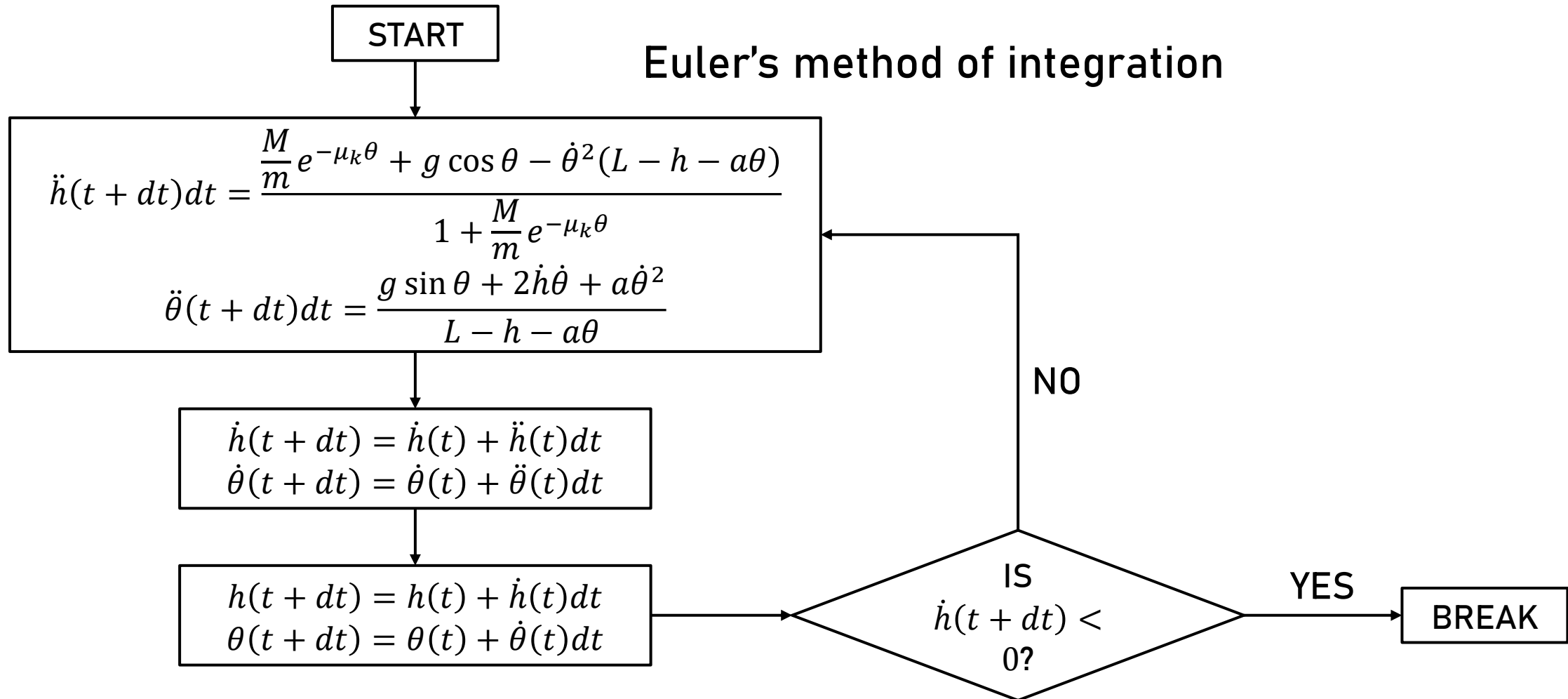


NUMERICAL SOLUTION

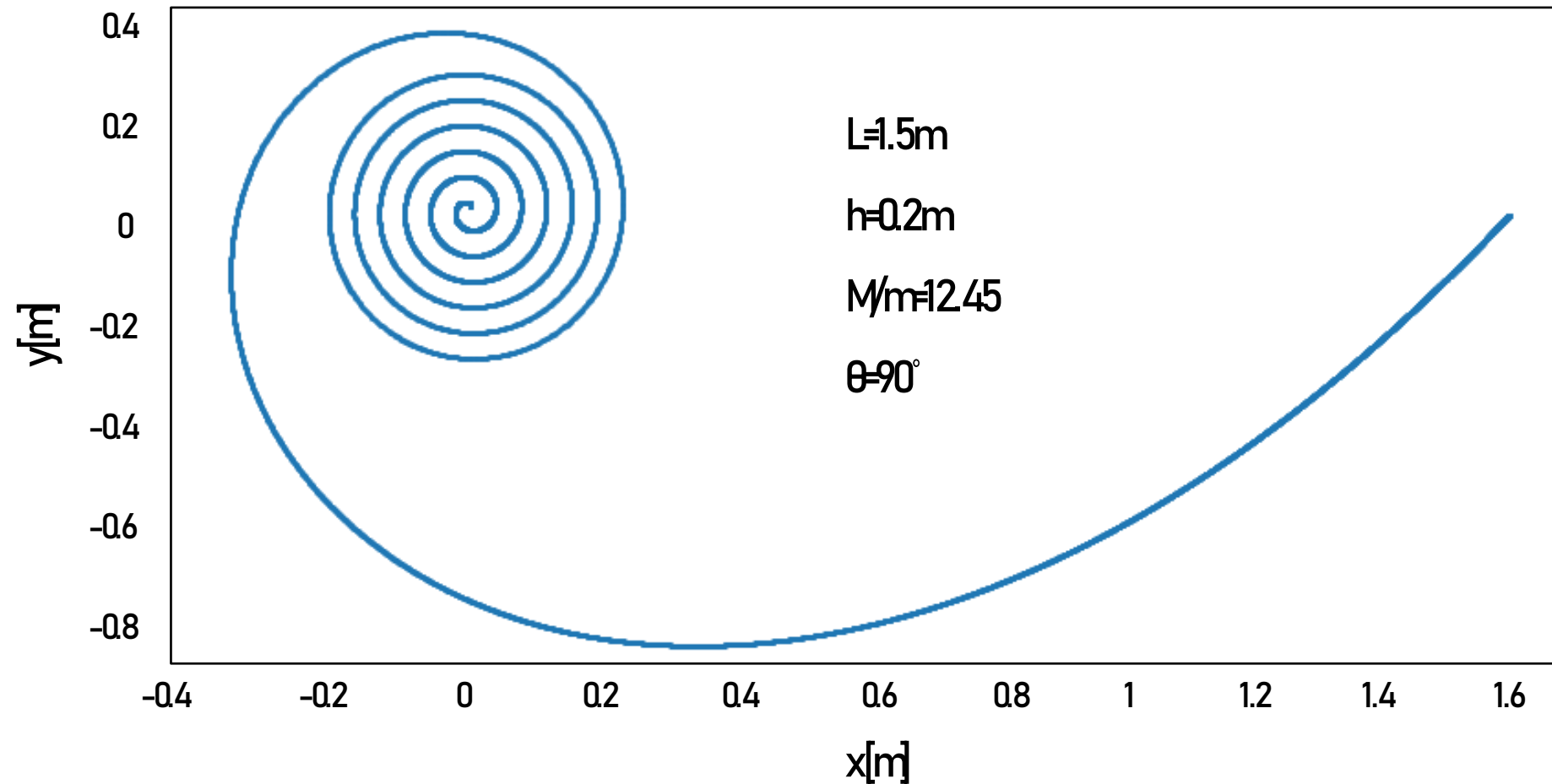


NUMERICAL SOLUTION_ ALGORITHM

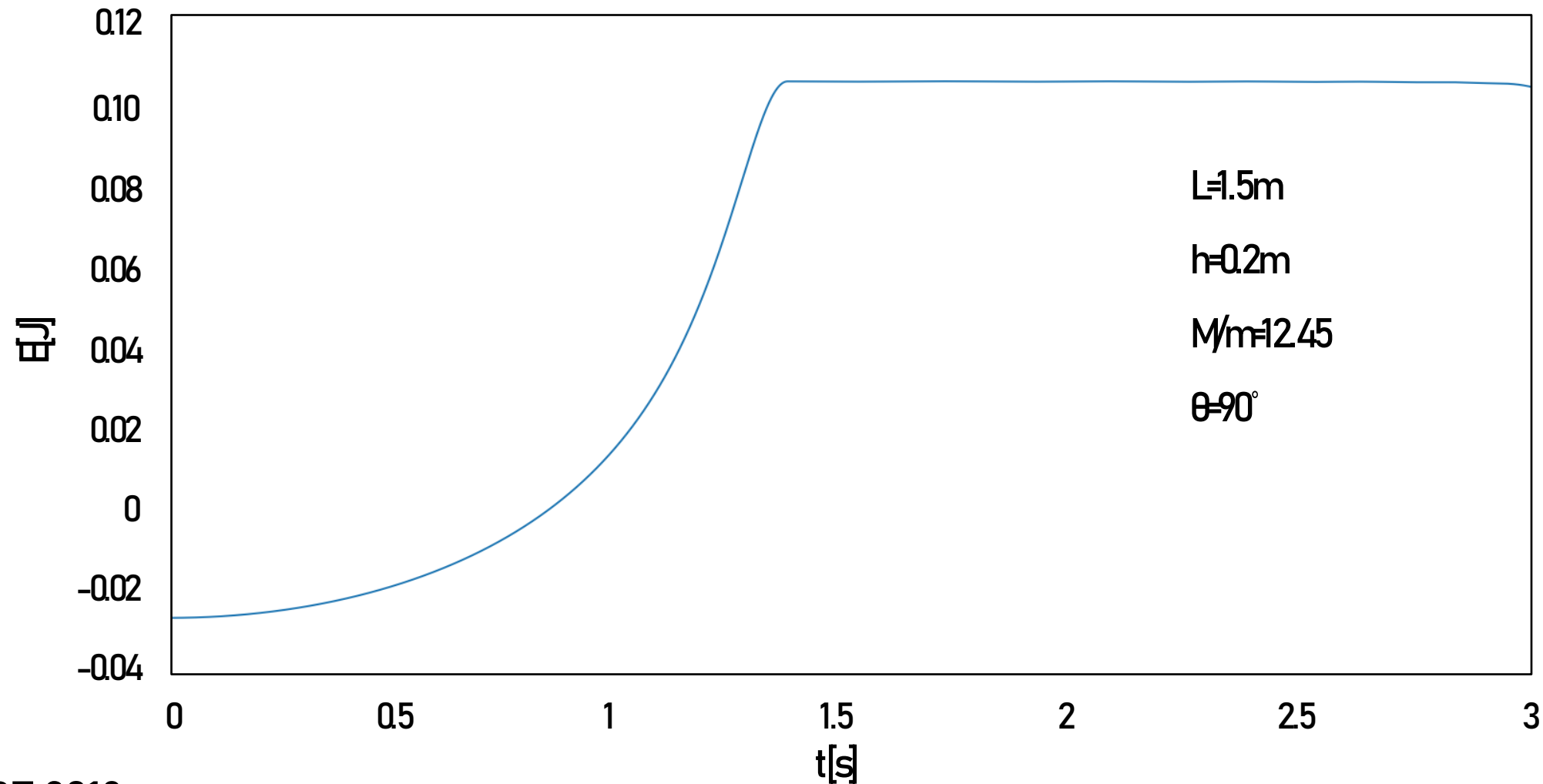
Euler's method of integration



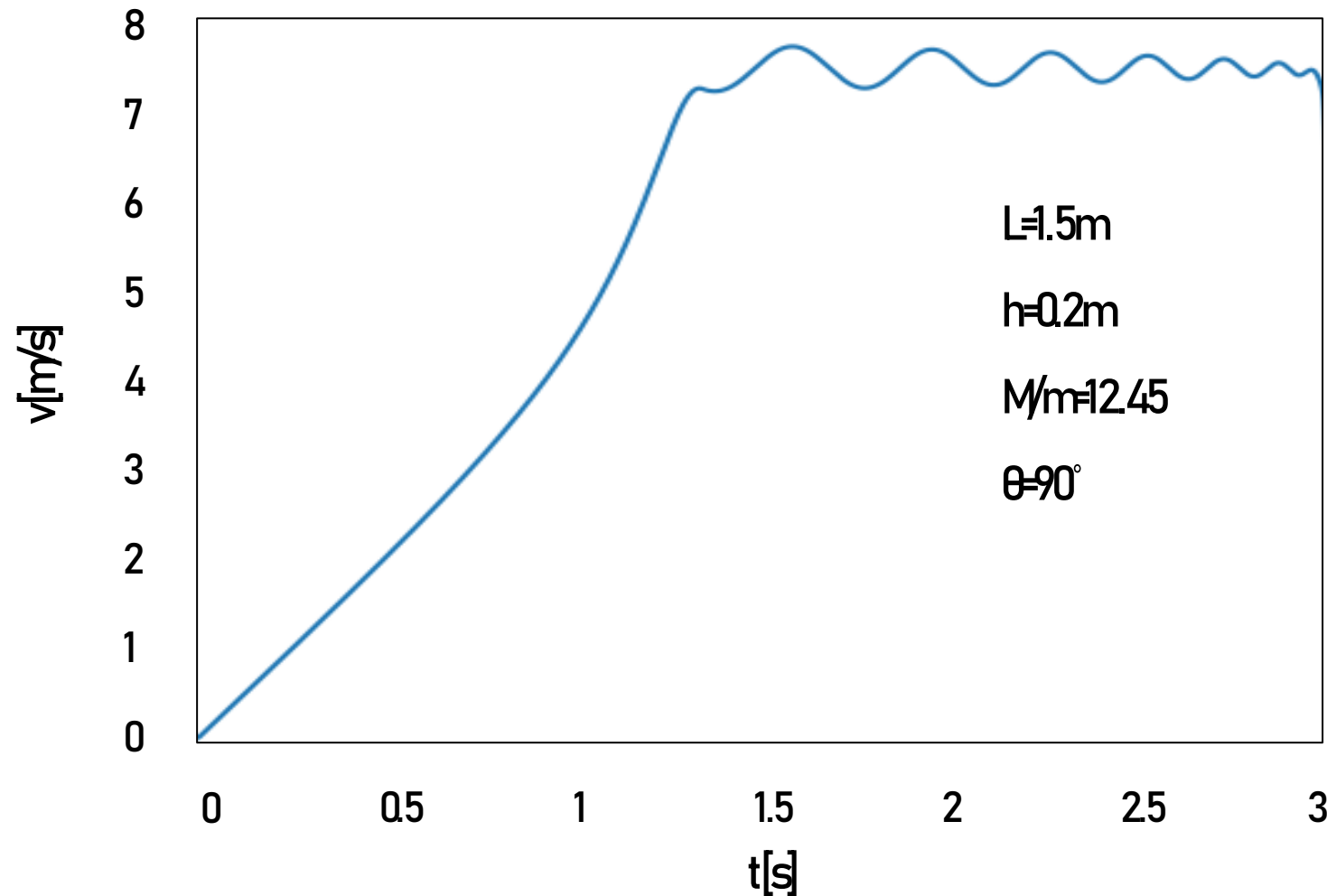
NUMERICAL SOLUTION_TRAJECTORIES

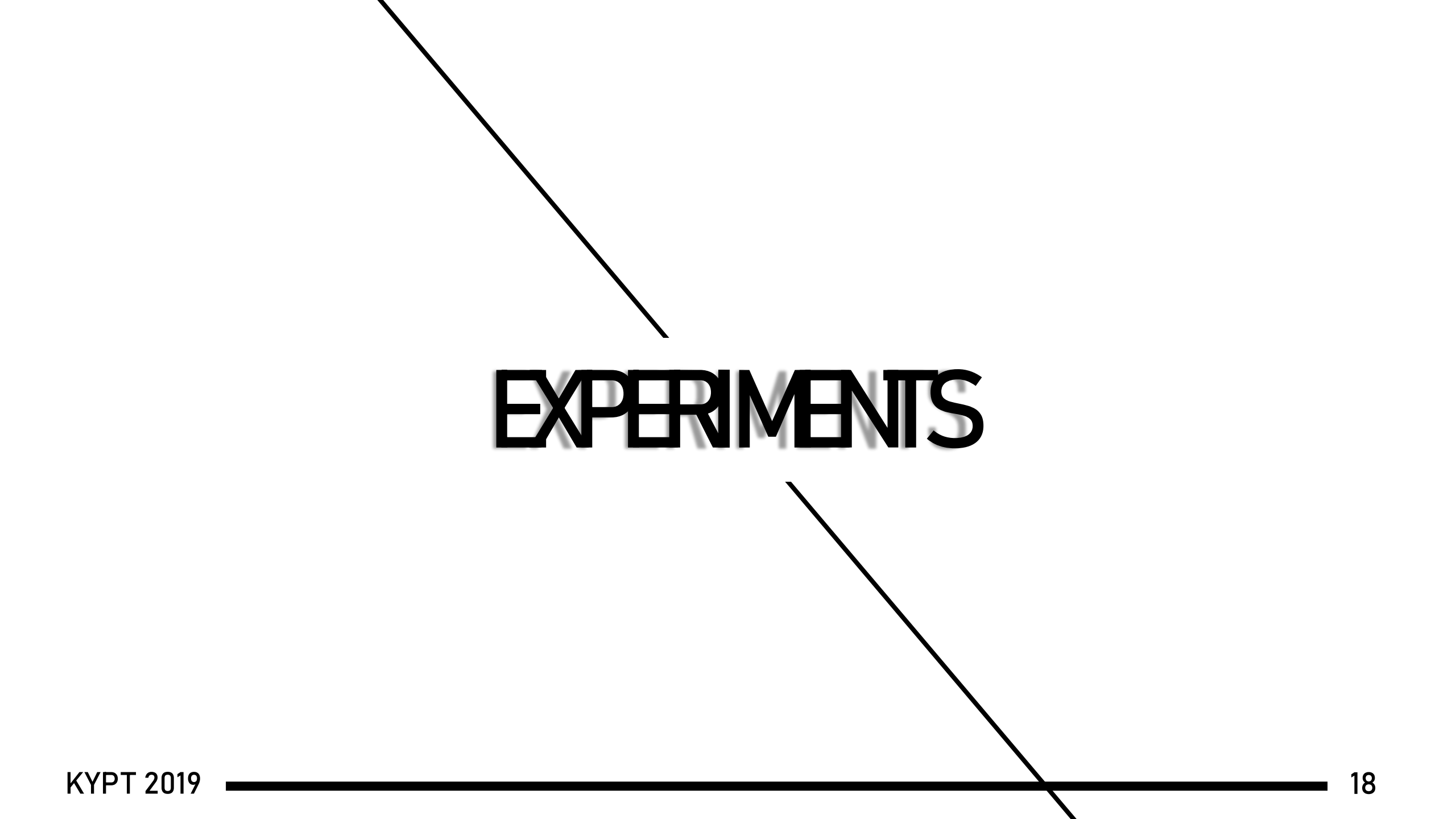


NUMERICAL SOLUTION_ ENERGY



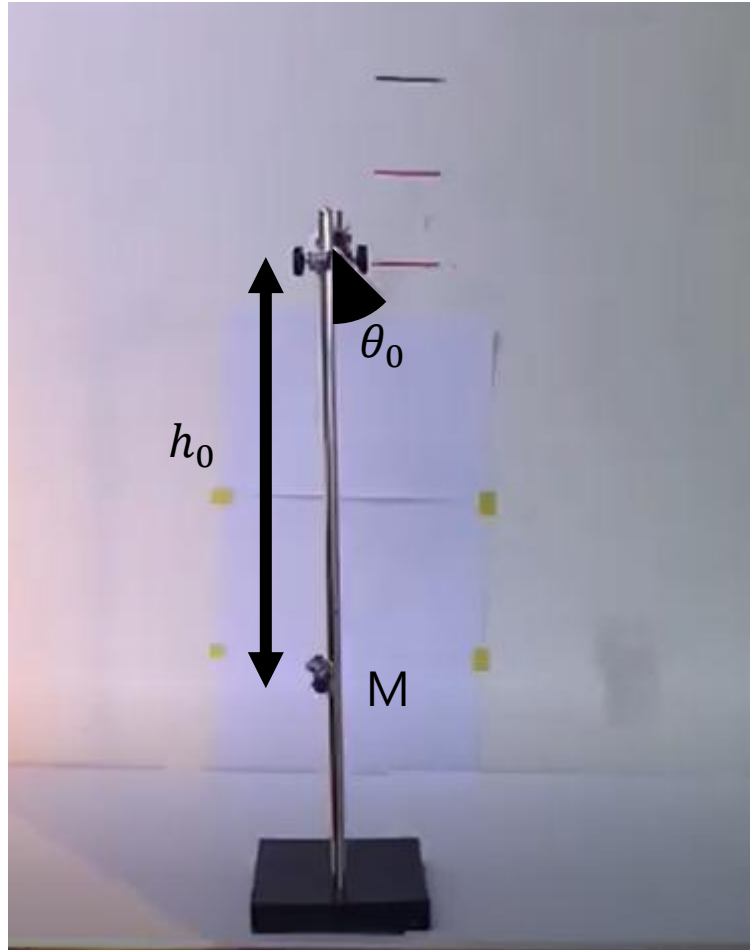
NUMERICAL SOLUTION_ VELOCITY





EXPERIMENTS

EXPERIMENT_SETTINGS



Control Variables

L = length of the thread

h_0 = length of the thread hanging

θ_0 = initial angle

m = mass of the small object

a = radius of axis

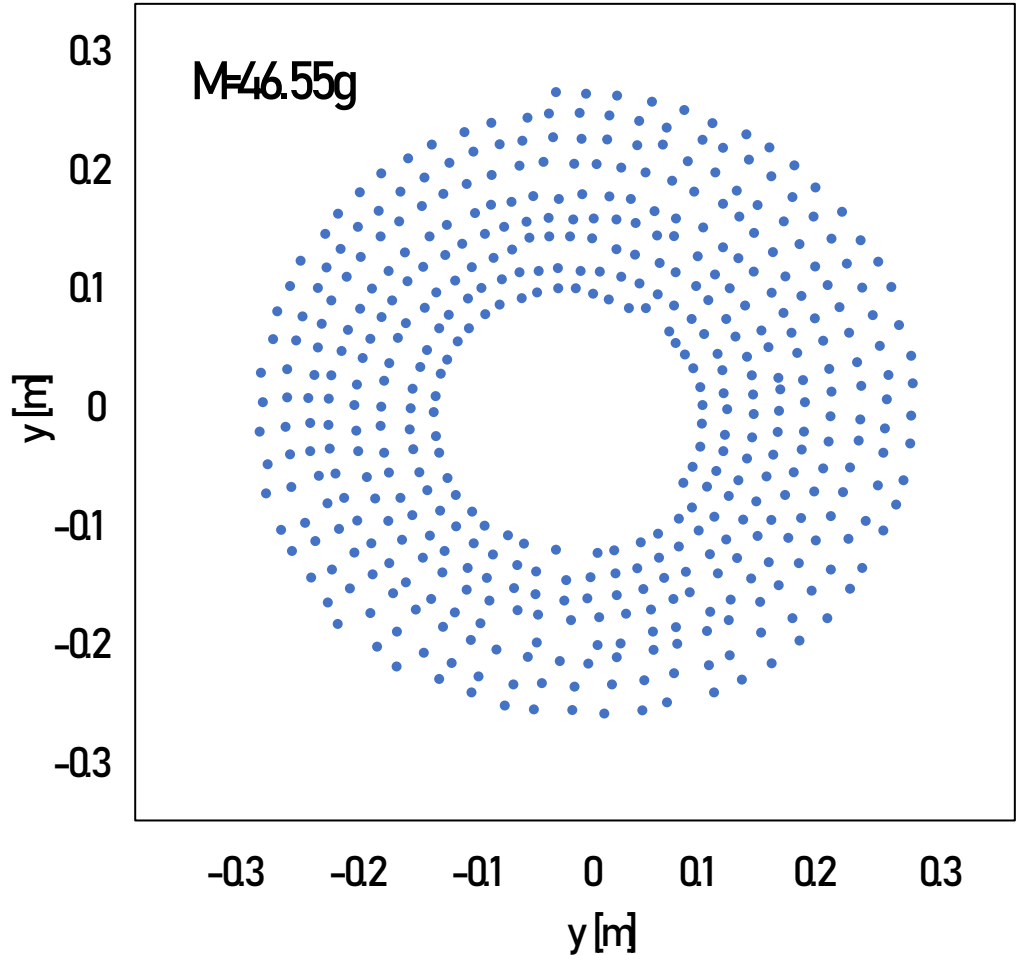
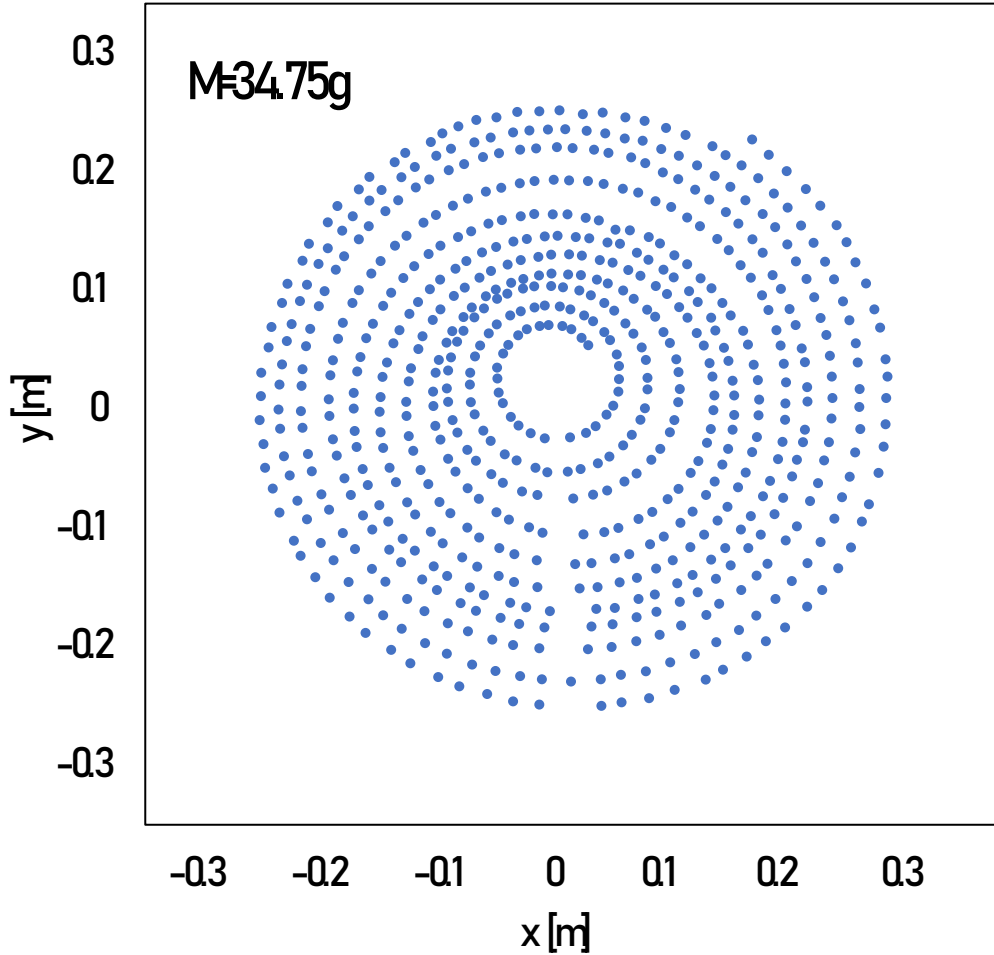
$114.30 \pm 0.01 \text{cm}$
$40.00 \pm 0.01 \text{cm}$
$45.0 \pm 0.1^\circ$
$4.21 \pm 0.01 \text{g}$
$0.30 \pm 0.01 \text{cm}$

Manipulated Variables

M = mass of the heavy object

$29.14 \pm 0.01 \text{g}$
$34.75 \pm 0.01 \text{g}$
$40.40 \pm 0.01 \text{g}$
$46.55 \pm 0.01 \text{g}$
$52.42 \pm 0.01 \text{g}$

EXPERIMENTS_TRAJECTORIES



L	$114.30 \pm 0.01 \text{cm}$
h_0	$40.00 \pm 0.01 \text{cm}$
θ_0	$45.0 \pm 0.1^\circ$
m	$4.21 \pm 0.01 \text{g}$
a	$0.30 \pm 0.01 \text{cm}$

EXPERIMENTS $E - t$

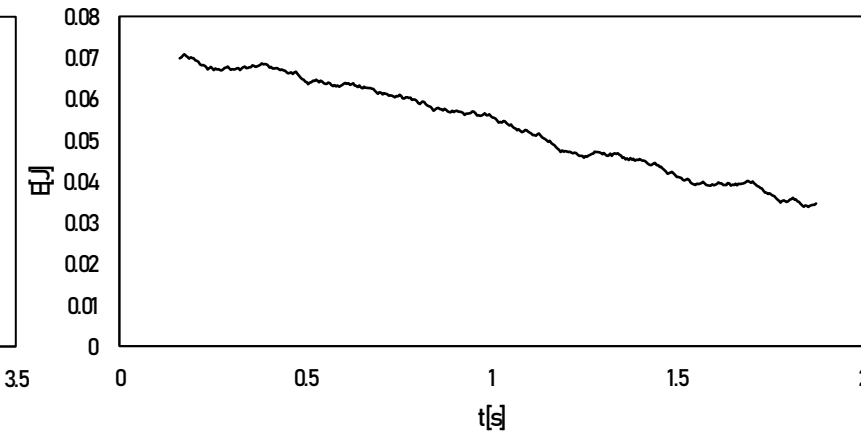
$M/m=6.92$



$M/m=8.25$



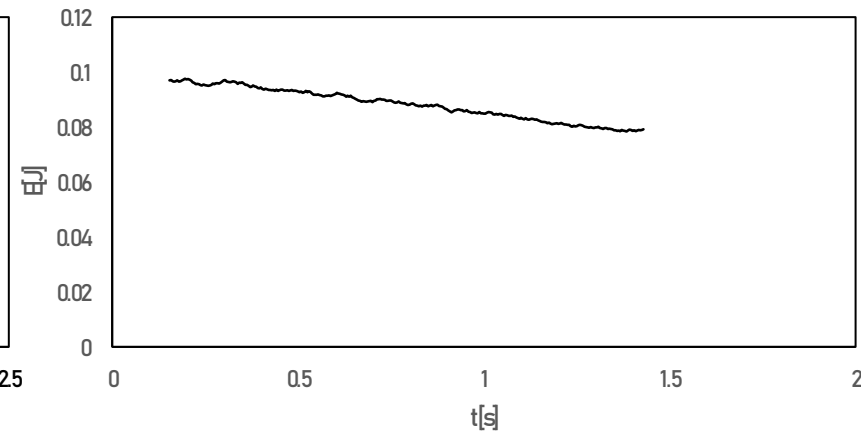
$M/m=9.60$



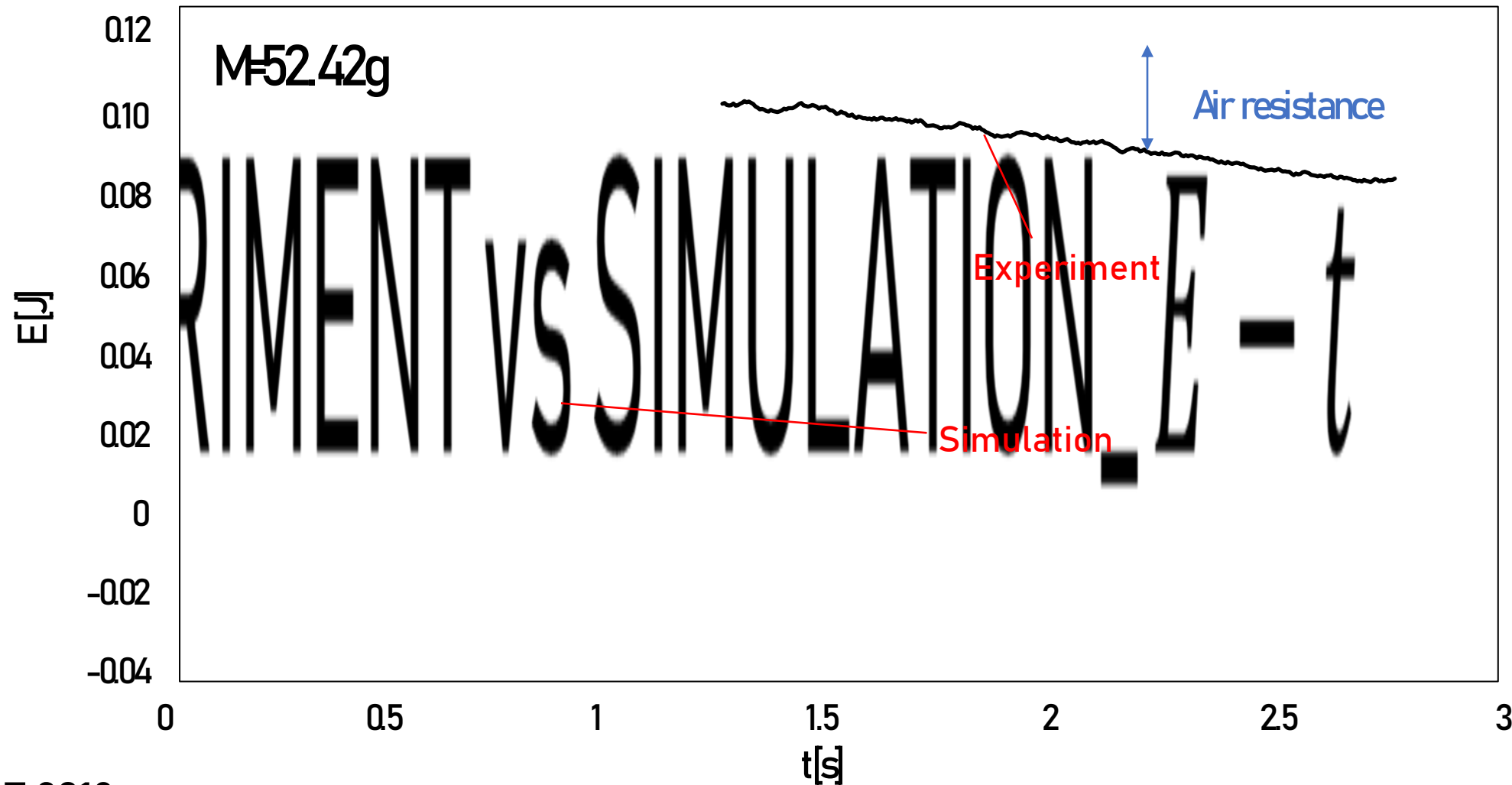
$M/m=11.06$



$M/m=12.45$

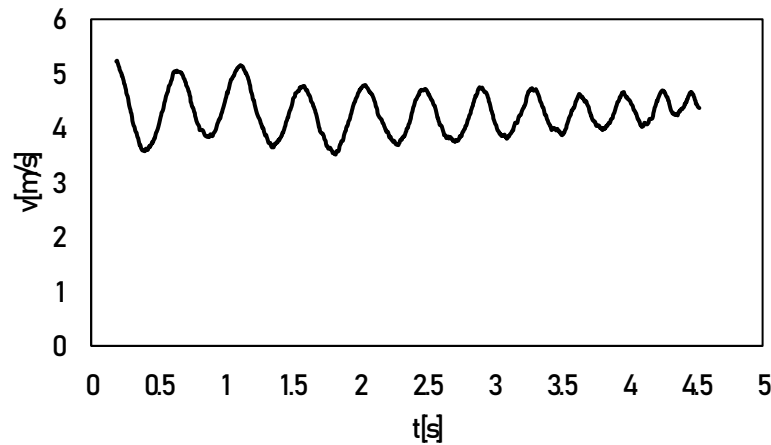


EXPERIMENT vs SIMULATION_ $E - t$

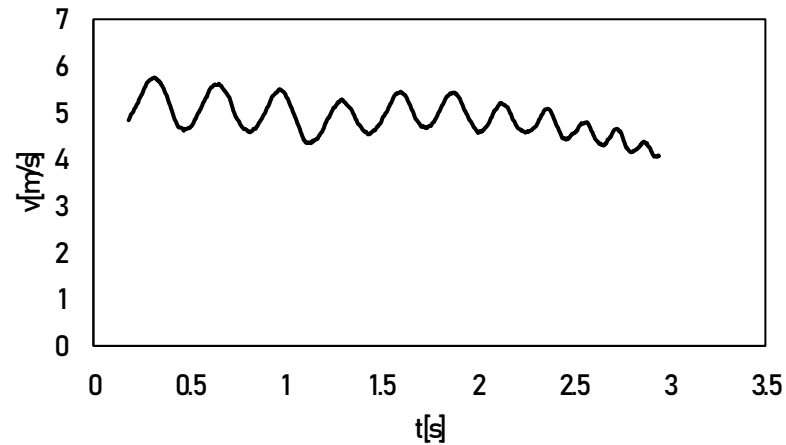


EXPERIMENTS_ $v - t$

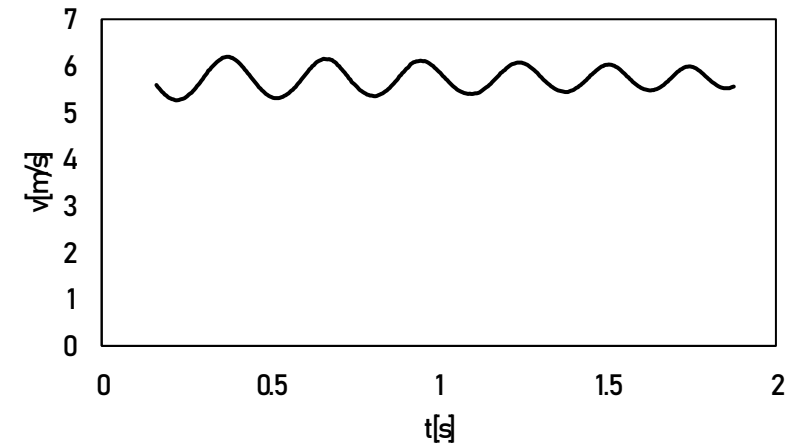
$M/m=6.92$



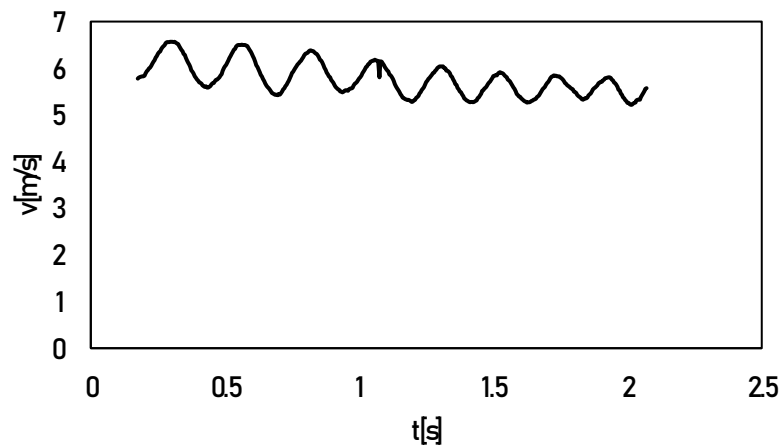
$M/m=8.25$



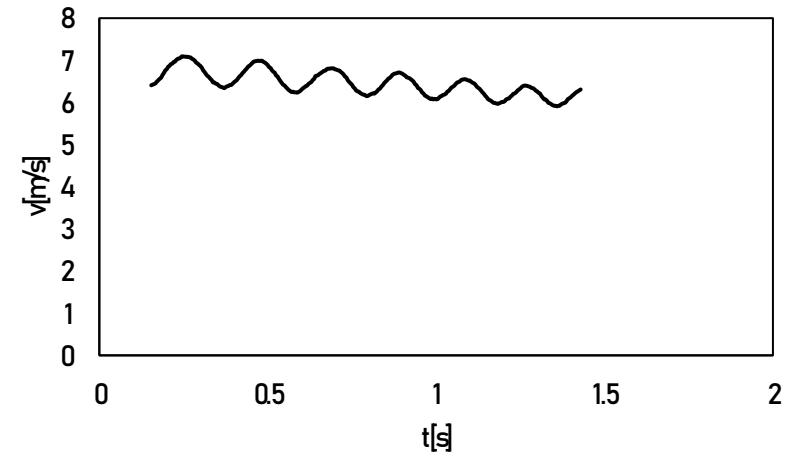
$M/m=9.60$



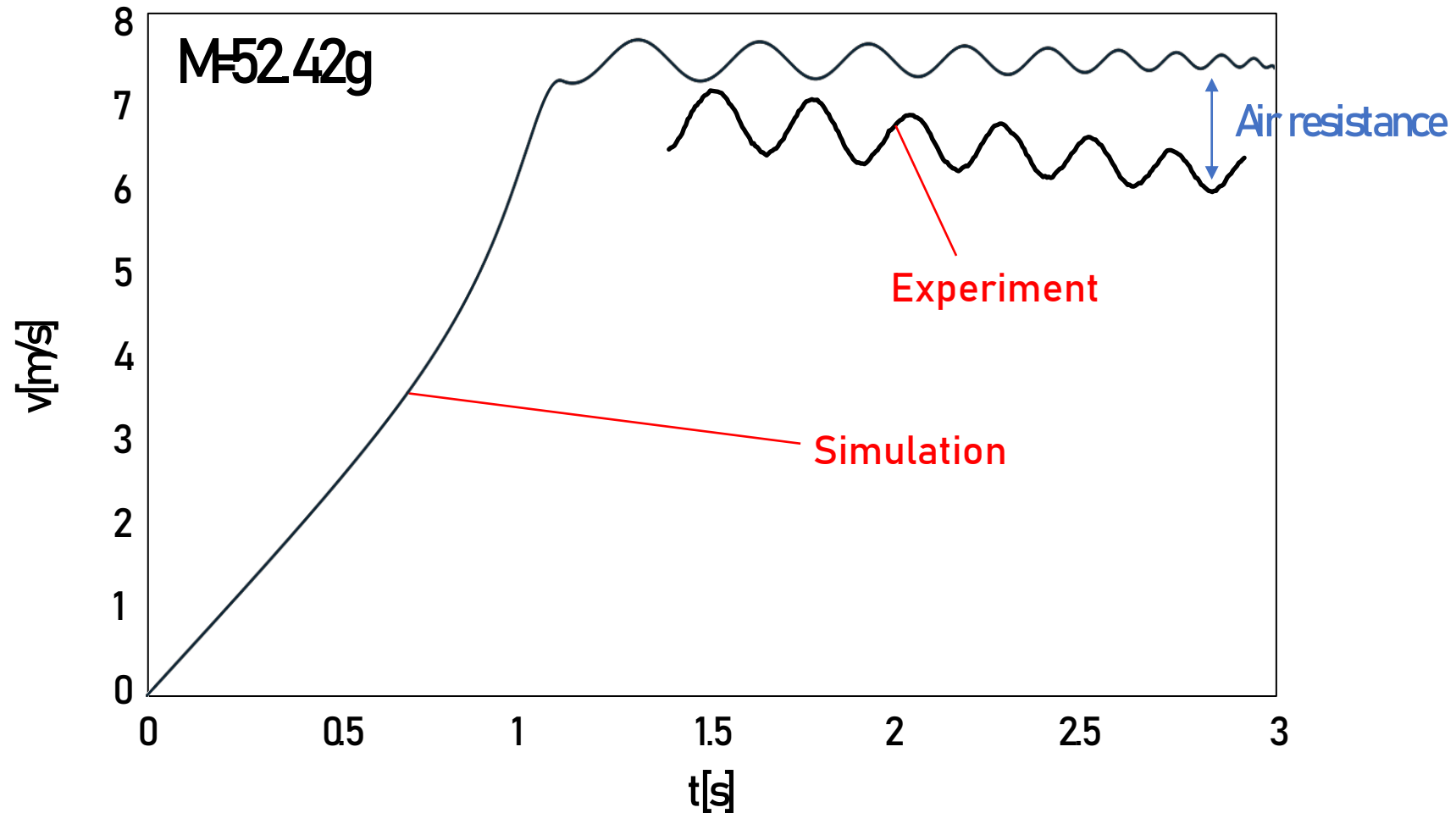
$M/m=11.06$



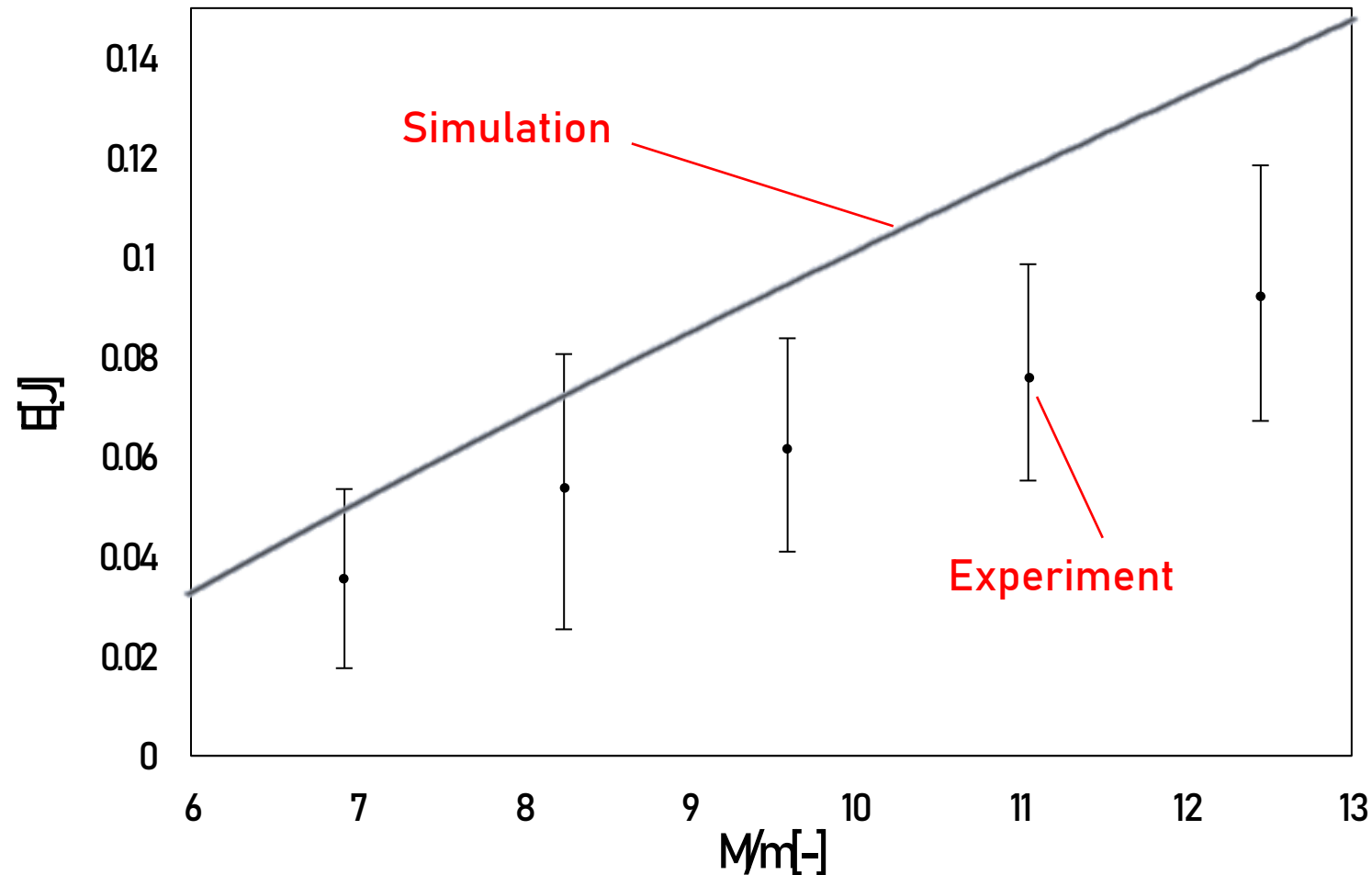
$M/m=12.45$

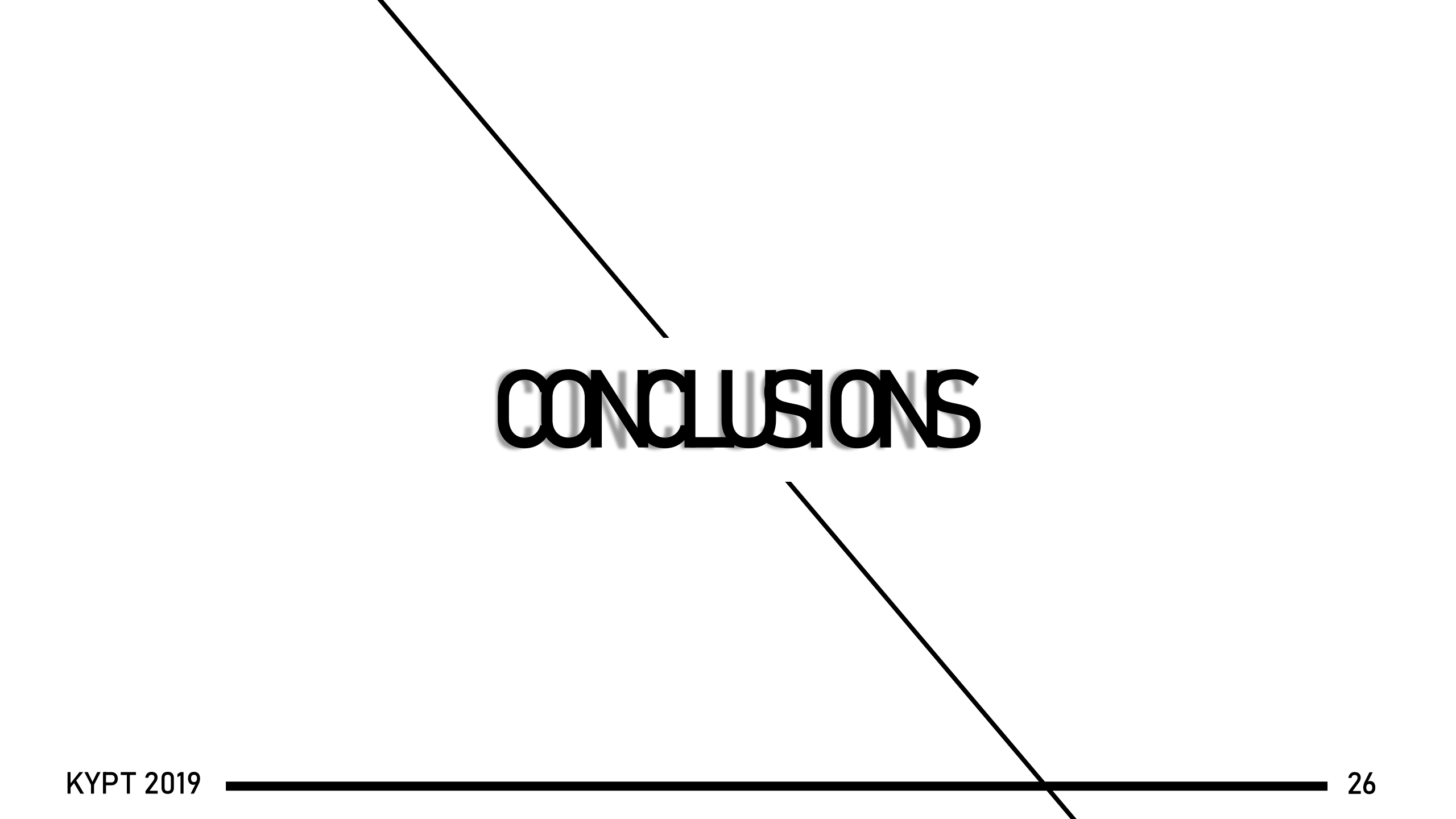


EXPERIMENT vs SIMULATION_ $v - t$



EXPERIMENT vs SIMULATION_ $E - M/m$

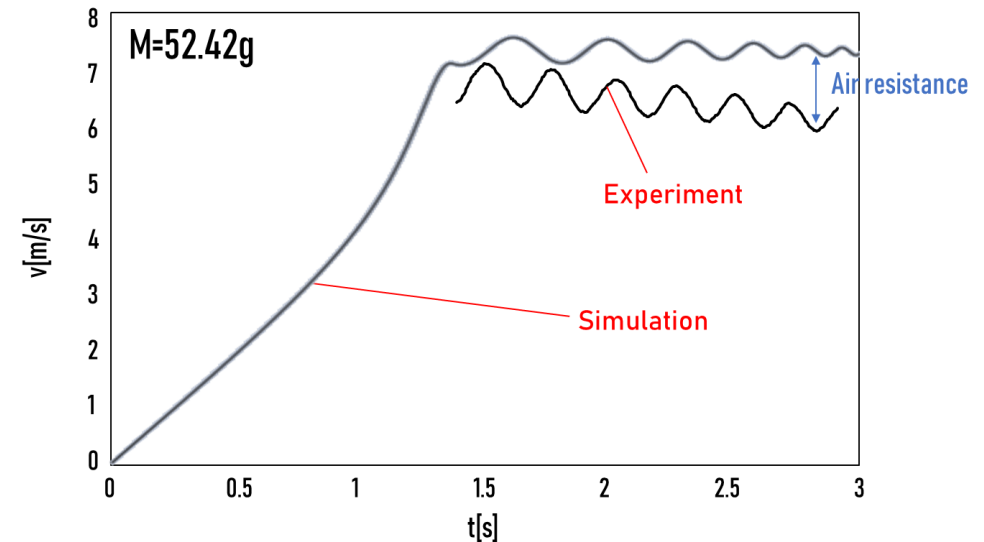
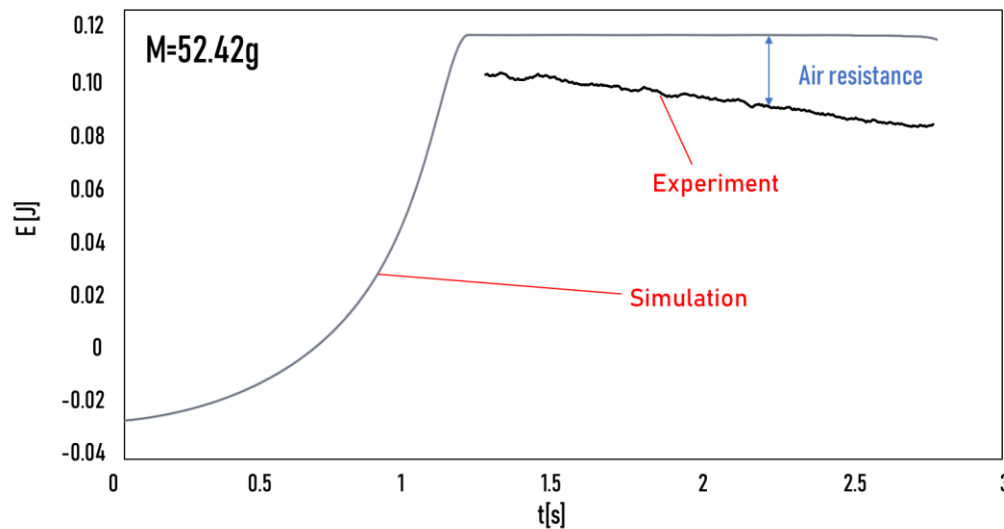


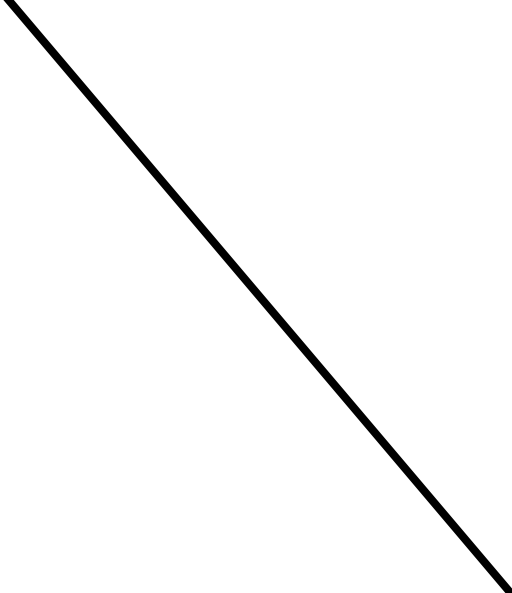


CONCLUSIONS

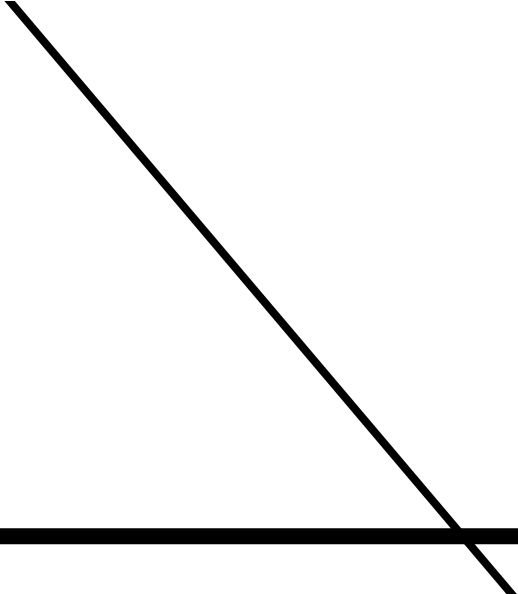
CONCLUSIONS

1. Energy is conserved throughout the phenomenon (with the neglect of air resistance).
2. The heavy load stops because of the increased tension.
3. Our numerical solutions accurately predict the motion of the objects.





APPENDIX



ASSUMPTIONS FOR CAPSTAN EQUATION

- The rope is on the verge of full sliding
- The line is not rigid
- The line is non-elastic.

————→ ASSUMPTION HOLDS!